

8-1

1. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$f(x) = \begin{cases} \sqrt{9-x^2} & \text{for } -3 \leq x \leq 0 \\ -x + 3\cos\left(\frac{\pi x}{2}\right) & \text{for } 0 < x \leq 4 \end{cases}$$

Let f be the function defined above.

- Find the average rate of change of f on the interval $-3 \leq x \leq 4$.
- Write an equation for the line tangent to the graph of f at $x = 3$.
- Find the average value of f on the interval $-3 \leq x \leq 4$.
- Must there be a value of x at which $f(x)$ attains an absolute maximum on the closed interval $-3 \leq x \leq 4$? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point does not require a simplified answer. Substitution of function values is required.



0	1
---	---

The student response accurately includes the criteria below.

☐ answer

Solution:

$$\text{Average rate of change} = \frac{f(4) - f(-3)}{4 - (-3)} = \frac{-1 - 0}{7} = -\frac{1}{7}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

8-1

The first point is earned with a correct and labeled value of $f'(3)$.

The second point can be earned with a consistent tangent line equation created from an incorrect value of $f'(3)$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $f'(3)$
- ☐ equation

Solution:

$$f(3) = -3 + 3 \cos\left(\frac{3\pi}{2}\right) = -3$$

$$\text{For } 0 < x < 4, f'(x) = -1 + \left(-3 \sin\left(\frac{\pi x}{2}\right)\right) \cdot \frac{\pi}{2}$$

$$f'(3) = -1 + \left(-3 \sin\left(\frac{3\pi}{2}\right)\right) \cdot \frac{\pi}{2} = -1 + \frac{3\pi}{2}$$

$$\text{An equation for the tangent line is } y = -3 + \left(-1 + \frac{3\pi}{2}\right)(x - 3).$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The constant $\frac{1}{b-a}$ is part of the fourth point.

Simplification is not required for the second point.

The third point is earned with the antiderivative $\frac{-x^2}{2} + \frac{6}{\pi} \sin\left(\frac{\pi x}{2}\right)$. The third point can also be earned with $\frac{-x^2}{2} + \frac{2}{\pi} \sin\left(\frac{\pi x}{2}\right)$, though this response will not be eligible for the fourth point.

Simplification is not required for the fourth point.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes **all four** of the criteria below.

- ☐ integrals of f over $-3 \leq x \leq 0$ and $0 \leq x \leq 4$

8-1

- ☐ value of $\int_{-3}^0 \sqrt{9-x^2} \, dx$
- ☐ antiderivative of $-x + 3 \cos\left(\frac{\pi x}{2}\right)$
- ☐ answer

Solution:

The average value of f on the interval $-3 \leq x \leq 4$ is $\frac{1}{4-(-3)} \int_{-3}^4 f(x) \, dx$.

$$\int_{-3}^4 f(x) \, dx = \int_{-3}^0 f(x) \, dx + \int_0^4 f(x) \, dx$$

$$\int_{-3}^0 f(x) \, dx = \int_{-3}^0 \sqrt{9-x^2} \, dx = \frac{9\pi}{4}$$

$$\int_0^4 f(x) \, dx = \int_0^4 \left(-x + 3 \cos\left(\frac{\pi x}{2}\right)\right) \, dx = \left[-\frac{1}{2}x^2 + \frac{6}{\pi} \sin\left(\frac{\pi x}{2}\right)\right]_0^4 = -8$$

$$\frac{1}{4-(-3)} \int_{-3}^4 f(x) \, dx = \frac{1}{7} \left(\frac{9\pi}{4} - 8\right)$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point can be earned by evaluating each branch of the function at $x = 0$ instead of evaluating the one-sided limits, but the response will not be eligible for the second point.

The second point is earned with correct limit notation for the first point, clear communication that $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0) = 3$, and a statement that the function is continuous on the closed interval, and therefore must have an absolute maximum. Specifically naming the Extreme Value Theorem is not required to earn this point. Responses that indicate that the function is differentiable on the interval $[-3, 4]$ will not earn the point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ continuity at $x = 0$
- ☐ answer with justification

8-1**Solution:**

$\lim_{x \rightarrow 0^-} f(x) = f(0) = 3$ and $\lim_{x \rightarrow 0^+} f(x) = 3$, so f is continuous at $x = 0$.

Because f is continuous on $[-3, 4]$, the Extreme Value Theorem guarantees that f attains an absolute maximum on $[-3, 4]$.

8-1

2. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The function f is given by $f(x) = \frac{x^3}{e^x}$.

- (a) Write an equation for the line tangent to the graph of f at $x = 5$.
- (b) Write an equation that involves a limit and demonstrates that the x -axis is an asymptote for the graph of f .
- (c) Find the x -coordinate of each critical point of f . Classify each critical point as the location of a relative minimum, a relative maximum, or neither. Justify your answers.
- (d) Find the average value of f' , the derivative of f , on the interval $[-1, 5]$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

$f'(x) = \frac{3x^2 - x^3}{e^x}$ earns the first 2 points.

An unlabeled correct $f'(x)$ earns the first 2 points.

To earn the first point:

- If the quotient rule was used, the presented derivative must include a sum or difference, a factor of e^x , and a polynomial in each term of the numerator.
- If the product rule was used, the presented derivative must include a factor of e^{-x} in each term.

A response that includes $f'(x) = \frac{e^x \cdot 3x^2 - x^3 e^x}{e^{x^2}}$ earns the first point, does not earn the second point, and is eligible for the third point. In this case the response must correctly evaluate this expression at $x = 5$, and use the resulting value as the slope of the tangent line.

The second point cannot be earned without the first. The second point is earned for a correct expression for $f'(x)$ or a correct value of $f'(5)$.

A response that labels the values $f(5)$ and $f'(5)$ correctly with no other supporting work does not earn the first 2 points but is eligible for the third point.

8-1

A response that presents a correct $f'(x)$ but an incorrect $f'(5)$ earns the first 2 points, but does not earn the third point.

A response that presents a correct equation of the tangent line with no supporting work earns no points in this part.

A response with any arithmetic error in $f'(x)$ does not earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Form of quotient rule (or product rule)
- ☐ $f'(x)$ or $f'(5)$
- ☐ Answer

Solution:

$$f'(x) = \frac{e^x \cdot 3x^2 - x^3 e^x}{(e^x)^2} = \frac{3x^2 - x^3}{e^x}$$

$$f'(5) = \frac{75 - 125}{e^5} = \frac{-50}{e^5}$$

$$f(5) = \frac{125}{e^5}$$

An equation for the tangent line is $y = \frac{125}{e^5} - \frac{50}{e^5}(x - 5)$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

$\lim_{x \rightarrow \infty} f(x) = 0$ or $\lim_{x \rightarrow \infty} \frac{x^3}{e^x} = 0$ earns the point.

A response that presents an expression, rather than an equation, does not earn the point.

A response that presents any additional incorrect limits does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

8-1

· Equation

Solution:

$$\lim_{x \rightarrow \infty} f(x) = 0$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

$(3x^2 - x^3) = 0$ earns the first point.

A response that presents only one of $x = 0$ or $x = 3$ along with correct answer and justification earns 1 of the last 2 points.

A response that lists additional values of x as critical points does not earn either of the last 2 points.

The second derivative may be used to justify the relative maximum at $x = 3$. Since $f''(x) = 0$ at $x = 0$, it is not sufficient to use the second derivative to conclude that $x = 0$ is neither a maximum or minimum value; more justification is needed.

A sign chart with no explanation is not sufficient justification.

Because the interval is not closed, the Extreme Value Theorem may not be used.

A response that imports e^{x^2} in the denominator of the derivative from part (a) is eligible for all 3 points.

A response that imports either $f'(x) = \frac{e^x \cdot 3x^2 + x^3 e^x}{(e^x)^2}$ or $f'(x) = \frac{-e^x \cdot 3x^2 + x^3 e^x}{(e^x)^2}$ from part (a) is eligible for all 3 points.

· An import of $f'(x) = \frac{e^x \cdot 3x^2 + x^3 e^x}{(e^x)^2}$ should find critical points $x = -3$ and $x = 0$ AND conclude that f has a relative minimum at $x = -3$ and neither a minimum nor a maximum at $x = 0$.

· An import of $f'(x) = \frac{-e^x \cdot 3x^2 + x^3 e^x}{(e^x)^2}$ should find critical points $x = 0$ and $x = 3$ AND conclude that f has neither a minimum nor a maximum at $x = 0$ and a relative minimum at $x = 3$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $f'(x) = 0$
 $x = 0$ and $x = 3$ only

8-1

☐☐

Answer with justification

Solution:

$$f'(x) = \frac{e^x \cdot 3x^2 - x^3 e^x}{(e^x)^2} = \frac{x^2 e^x (3-x)}{e^{2x}} = 0 \Rightarrow x = 0, x = 3$$

$f'(x)$ does not change sign at $x = 0$. Therefore, f has neither a minimum nor a maximum at $x = 0$.

$f'(x)$ changes sign from positive to negative at $x = 3$. Therefore, f has a relative maximum at $x = 3$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A factor of $\frac{1}{6}$ is not needed to earn the first point.

The definite integral can be presented in a single or multiple steps to earn the first point. For example, the following response earns both points: $\int f'(x) dx = f(x)$, so the average value is $\frac{1}{6}(\frac{125}{e^5} + \frac{1}{e^{-1}})$.

$f(5) - f(-1)$ earns the first point and $\frac{1}{6}(\frac{125}{e^5} - (\frac{-1}{e^{-1}}))$ earns both points, provided there are no subsequent errors in simplification.

A response that presents a correct integral along with the correct average value, but provides incorrect or incomplete communication, earns 1 out of 2 points. For example, the following response earns 1 out of 2 points:

$$\int_{-1}^5 f'(x) dx = (\frac{125}{e^5} + e) = \frac{1}{6}(\frac{125}{e^5} + e).$$

$\int_{-1}^5 f'(x) dx$ with an incorrect $f'(x)$ imported from part (a) or part (c) earns the first point.

A response with incorrect limits earns no points.

A response of $\frac{1}{6} \cdot \frac{125}{e^5} - \frac{-1}{e^{-1}}$ or $\frac{1}{6} \cdot f(5) - f(1)$ earns the first point but not the second point.

A response that uses an incorrect value of $f(5)$ or $f(-1)$ does not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

8-1

- ☐ Integral
- ☐ Answer

Solution:

$$\frac{1}{5-(-1)} \int_{-1}^5 f'(x) dx$$

$$= \frac{1}{6}(f(5) - f(-1)) = \frac{1}{6}\left(\frac{125}{e^5} - \left(\frac{-1}{e^{-1}}\right)\right) = \frac{1}{6}\left(\frac{125}{e^5} + e\right)$$

3.  The rate at which water is sprayed on a field of vegetables is given by $R(t) = 2\sqrt{1 + 5t^3}$, where t is in minutes and $R(t)$ is in gallons per minute. During the time interval $0 \leq t \leq 4$, what is the average rate of water flow, in gallons per minute?

- (A) 8.458
- (B) 13.395
- (C) 14.691
- (D) 18.916
- (E) 35.833



4. If the average value of the function f over the closed interval $[2, 4]$ is 3 and if $f(x) \geq 0$ for all x in $[2, 4]$, what is the area of the region enclosed by the graph of $y = f(x)$, the lines $x = 2$ and $x = 4$, and the x -axis?

- (A) 12
- (B) 6
- (C) 3
- (D) $3/2$



5.  If $f(x) = (x + 2) \sin(\sqrt{x + 2})$, what is the average value of f on the closed interval $[0, 6]$?

- (A) 2.220
- (B) 3.348
- (C) 4.757
- (D) 20.090
- (E) 28.541



8-1

6. Let f be the function defined by $f(x) = \frac{1}{x}$. What is the average value of f on the interval $[4, 6]$?
- (A) $-1/24$
(B) $5/24$
(C) $(1/2) \ln(3/2)$ ✓
(D) $\ln(3/2)$
(E) $(1/2) \ln 2$
7. If the average value of a continuous function f on the interval $[-2, 4]$ is 12, what is $\int_{-2}^4 \frac{f(x)}{8} dx$?
- (A) $\frac{3}{2}$
(B) 3
(C) 9 ✓
(D) 72
8. For time $t \geq 0$, the height h of an object suspended from a spring is given by $h(t) = 16 + 7 \cos\left(\frac{\pi t}{4}\right)$. What is the average height of the object from $t = 0$ to $t = 2$?
- (A) 16
(B) $\frac{39}{2}$
(C) $16 - \frac{14}{\pi}$
(D) $16 + \frac{14}{\pi}$ ✓
(E) $32 + \frac{28}{\pi}$
9. If the average value of a continuous function f on the interval $[-2, 4]$ is 12, what is $\int_{-2}^4 \frac{f(x)}{8} dx$?
- (A) $\frac{3}{2}$
(B) 3
(C) 9 ✓
(D) 72

8-1

10. Let f be a continuous function for all real numbers. Let g be the function defined by $g(x) = \int_1^x f(t)dt$.

If the average rate of change of g on the interval $2 \leq x \leq 5$ is 6, which of the following statements must be true?

(A) The average value of f on the interval $2 \leq x \leq 5$ is 6. ✓

(B) $g'(2) = 6$

(C) $\frac{g'(5)+g'(2)}{2} = 6$

(D) $\int_2^5 g(x)dx = 6$

11.  What is the average value of $y = \sqrt{\cos x}$ on the interval $0 \leq x \leq \frac{\pi}{2}$?

(A) -0.637

(B) 0.500

(C) 0.763 ✓

(D) 1.198

(E) 1.882

12.  What is the average value of $y = \tan\left(\frac{x^2}{9}\right)$ on the closed interval $[1.25, 2]$?

(A) 0.116

(B) 0.232

(C) 0.310 ✓

(D) 0.326

Answer C

This option is correct. The average value of a function f over the interval $a \leq x \leq b$ is $\frac{1}{b-a} \int_a^b f(x) dx$.

Applying that formula to this problem gives the calculation:

$$\frac{\int_{1.25}^2 \tan\left(\frac{x^2}{9}\right) dx}{2-1.25} = 0.310$$

8-1

13.

Time (weeks)	0	2	6	10
Level	210	200	190	180

The table above gives the level of a person's cholesterol at different times during a 10-week treatment period.

What is the average level over this 10-week period obtained by using a trapezoidal approximation with the subintervals $[0, 2]$, $[2, 6]$, and $[6, 10]$?

(A) 188

(B) 193

(C) 195

(D) 198



14.

 The temperature F , in degrees Fahrenheit ($^{\circ}\text{F}$), of a cup of coffee t minutes after it is poured is given by $F(t) = 72 + 118e^{-0.093t}$. To the nearest degree, what is the average temperature of the coffee between $t = 0$ and $t = 10$ minutes?

(A) 93°F (B) 119°F (C) 146°F (D) 149°F (E) 154°F 

15.

The velocity of a particle moving along the x -axis is given by $v(t) = 2 - t^2$ for time $t > 0$. What is the average velocity of the particle from time $t = 1$ to time $t = 3$?

(A) -4

(B) -3

(C) $-\frac{7}{3}$ (D) $\frac{7}{3}$ 

Answer C

This option is correct. The average value of $v(t)$ over the interval $[1, 3]$ is computed with a definite integral:

$$\begin{aligned}\frac{1}{3-1} \int_1^3 v(t) dt &= \frac{1}{3-1} \int_1^3 (2 - t^2) dt = \frac{1}{2} \left(2t - \frac{1}{3}t^3 \right) \Big|_1^3 \\ &= \frac{1}{2} \left((6 - 9) - \left(2 - \frac{1}{3} \right) \right) = -\frac{7}{3}.\end{aligned}$$

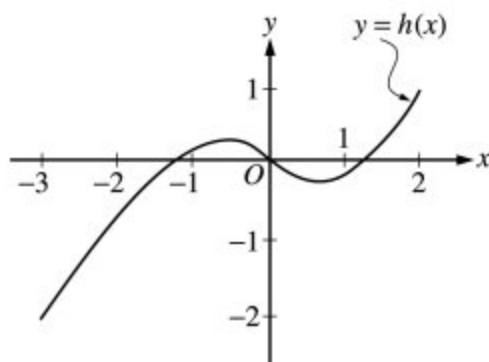
8-1

16.  On a certain day, the temperature, in degrees Fahrenheit, in a small town t hours after midnight ($t = 0$) is modeled by the function $g(t) = 65 - 8 \sin\left(\frac{\pi t}{12}\right)$. What is the average temperature in the town between 3 A.M. ($t = 3$) and 6 A.M. ($t = 6$), in degrees Fahrenheit?

- (A) 57.609
(B) 57.797
(C) 58.172
(D) 59.907

(B) 57.797 ✓

17.



The graph of the function h is shown in the figure above. Of the following, which has the greatest value?

- (A) Average value of h over
(B) Average rate of change of h over $[-3, 2]$
(C) $\int_{-3}^2 h(x) dx$
(D) $\int_{-3}^0 h(x) dx$
(E) $h'(0)$

(B) Average rate of change of h over $[-3, 2]$ ✓

8-1

18.

t (hours)	0	0.4	0.8	1.2	1.6	2.0	2.4
$v(t)$ (miles per hour)	0	11.8	9.5	17.2	16.3	16.8	20.1

Ruth rode her bicycle on a straight trail. She recorded her velocity $v(t)$, in miles per hour, for selected values of t over the interval $0 \leq t \leq 2.4$, as shown in the table above. For $0 < t \leq 2.4$, $v(t) > 0$.



For $0 \leq t \leq 4$ hours, Ruth's velocity can be modeled by the function g given by $g(t) = \frac{24t + 5 \sin(6t)}{t + 0.7}$. According to the model, what was Ruth's average velocity during the time interval $0 \leq t \leq 4$?

Part C

The response can earn up to 2 points:

1 point: For the correct integral

$$\text{Average velocity} = \frac{1}{2.4} \int_0^{2.4} g(t) dt$$

1 point: for the correct answer

Average velocity = 14.064 miles/hr



0	1	2
---	---	---

The response earns both of the following points:

1 point: For the correct integral

$$\text{Average velocity} = \frac{1}{2.4} \int_0^{2.4} g(t) dt$$

1 point: for the correct answer

Average velocity = 14.064 miles/hr

8-1

19.



t (hours)	0	0.4	0.8	1.2	1.6	2.0	2.4
$v(t)$ (miles per hour)	0	11.8	9.5	17.2	16.3	16.8	20.1

Ruth rode her bicycle on a straight trail. She recorded her velocity $v(t)$, in miles per hour, for selected values of t over the interval $0 \leq t \leq 2.4$ hours, as shown in the table above. For $0 < t \leq 2.4$, $v(t) > 0$.

(a) Use the data in the table to approximate Ruth's acceleration at time $t = 1.4$ hours. Show the computations that lead to your answer. Indicate units of measure.

(b) Using correct units, interpret the meaning of $\int_0^{2.4} v(t) \, dt$ in the context of the problem. Approximate $\int_0^{2.4} v(t) \, dt$ using a midpoint Riemann sum with three subintervals of equal length and values from the table.

(c) For $0 \leq t \leq 2.4$ hours, Ruth's velocity can be modeled by the function g given by

$g(t) = \frac{24t + 5 \sin(6t)}{t + 0.7}$. According to the model, what was Ruth's average velocity during the time interval $0 \leq t \leq 2.4$?

(d) According to the model, $g(t) = \frac{24t + 5 \sin(6t)}{t + 0.7}$, is Ruth's speed increasing or decreasing at time $t = 1.3$? Give a reason for your answer.

Part A

The response can earn up to 2 points:

1 point is earned for correct approximation

1 point is earned for correct units

$$a(1.4) \approx \frac{v(1.6) - v(1.2)}{1.6 - 1.2} = \frac{16.3 - 17.2}{1.6 - 1.2} = -2.25 \text{ miles/hr}^2$$



0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for correct approximation

1 point is earned for correct units

$$a(1.4) \approx \frac{v(1.6) - v(1.2)}{1.6 - 1.2} = \frac{16.3 - 17.2}{1.6 - 1.2} = -2.25 \text{ miles/hr}^2$$

8-1

Part B

The response can earn up to 3 points:

1 point is earned for the correct interpretation

$\int_0^{2.4} v(t) dt$ is the total distance Ruth traveled, in miles, from time $t = 0$ to time $t = 2.4$ hours.

1 point is earned for use of the midpoint Riemann sum

1 point is earned for the correct approximation

$$\begin{aligned}\int_0^{2.4} v(t) dt &\approx (0.8)(11.8) + (0.8)(17.2) + (0.8)(16.8) \\ &= 36.64 \text{ miles}\end{aligned}$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point is earned for the correct interpretation

$\int_0^{2.4} v(t) dt$ is the total distance Ruth traveled, in miles, from time $t = 0$ to time $t = 2.4$ hours.

1 point is earned for use of the midpoint Riemann sum

1 point is earned for the correct approximation

$$\begin{aligned}\int_0^{2.4} v(t) dt &\approx (0.8)(11.8) + (0.8)(17.2) + (0.8)(16.8) \\ &= 36.64 \text{ miles}\end{aligned}$$

Part C

The response can earn up to 2 points:

1 point is earned for the correct integral

$$\text{Average velocity} = \frac{1}{2.4} \int_0^{2.4} g(t) dt$$

1 point is earned for the correct answer

8-1

Average velocity = 14.064 miles/hr



0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for the correct integral

$$\text{Average velocity} = \frac{1}{2.4} \int_0^{2.4} g(t) dt$$

1 point is earned for the correct answer

Average velocity = 14.064 miles/hr

Part D

The response can earn up to 2 points:

2 points are earned for correct conclusion with reason

$$\text{Velocity} = g(1.3) = 18.096358 > 0$$

$$\text{Acceleration} = g'(1.3) = 3.761152 > 0$$

Ruth's speed is increasing at time $t = 1.3$ since velocity and acceleration have the same sign at this time.



0	1	2
---	---	---

The response earns all of the following points:

2 points are earned for correct conclusion with reason

$$\text{Velocity} = g(1.3) = 18.096358 > 0$$

$$\text{Acceleration} = g'(1.3) = 3.761152 > 0$$

Ruth's speed is increasing at time $t = 1.3$ since velocity and acceleration have the same sign at this time.

8-1

20. The following are related to this scenario:

A store is having a 12-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function S defined by $S(t) = 0.5t^4 - 16t^3 + 144t^2$ for $0 \leq t \leq 12$. At time $t = 0$, when the sale begins, there are no shoppers in the store.

 Find the value of $\frac{1}{3} \int_6^9 S'(t) dt$. Using correct units, explain the meaning of $\frac{1}{3} \int_6^9 S'(t) dt$ in the context of this problem.

Part B

The response can earn up to 2 points:

1 point: For the correct value of $\frac{1}{3} \int_6^9 S'(t) dt$

$$\begin{aligned} \frac{1}{3} \int_6^9 S'(t) dt &= \frac{1}{3} (S(9) - S(6)) \\ &= \frac{1}{3} (3280.5 - 2376) = 301.5 \end{aligned}$$

1 point: For the meaning of $\frac{1}{3} \int_6^9 S'(t) dt$

$\frac{1}{3} \int_6^9 S'(t) dt$ is the average rate of which the shoppers are entering the store in shoppers/hr between times $t = 6$ and $t = 9$ hours.



0	1	2
---	---	---

The response earns both of the following points:

1 point: For the correct value of $\frac{1}{3} \int_6^9 S'(t) dt$

$$\begin{aligned} \frac{1}{3} \int_6^9 S'(t) dt &= \frac{1}{3} (S(9) - S(6)) \\ &= \frac{1}{3} (3280.5 - 2376) = 301.5 \end{aligned}$$

1 point: For the meaning of $\frac{1}{3} \int_6^9 S'(t) dt$

8-1

$\frac{1}{3} \int_6^9 S'(t) dt$ is the average rate of which the shoppers are entering the store in shoppers/hr between times $t = 6$ and $t = 9$ hours.

8-1

21. The penguin population on an island is modeled by a differentiable function P of time t , where $P(t)$ is the number of penguins and t is measured in years for $0 \leq t \leq 40$. There are 100,000 penguins on the island at time $t = 0$. The birth rate for the penguins on the island is modeled by

$$B(t) = 1000e^{0.06t} \text{ penguins per year}$$

and the death rate for the penguins on the island is modeled by

$$D(t) = 250e^{0.1t} \text{ penguins per year.}$$



To the nearest whole number, what is the average rate of change of the penguin population on the island for $0 \leq t \leq 40$?

Part C

The response can earn up to 1 point:

1 point: For the correct answer

$$\frac{1}{40} \int_0^{40} (B(t) - D(t)) dt = 826.439$$

OR

$$\frac{P(40) - P(0)}{40 - 0} = \frac{133058 - 100000}{40} = 826.45$$

The average rate of change is 826 penguins per year.



0	1
---	---

The response earns one of the following point:

1 point: For the correct answer

$$\frac{1}{40} \int_0^{40} (B(t) - D(t)) dt = 826.439$$

OR

$$\frac{P(40) - P(0)}{40 - 0} = \frac{133058 - 100000}{40} = 826.45$$

The average rate of change is 826 penguins per year.

8-1

22. Find the value of c for which the average value of f over the interval $-5 \leq x \leq c$ is $\frac{1}{2}$.

Enter Scoring Guidelines Here

1 point is earned for integral

1 point is earned for equation

1 point is earned for answer

$$\frac{1}{c+5} \int_{-5}^c f(x) dx = \frac{1}{2}$$

$$\frac{1}{c+5}(-\pi + (-1) + 2 + c) = \frac{1}{2}$$

$$c = 3 + 2\pi$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for integral

1 point is earned for equation

1 point is earned for answer

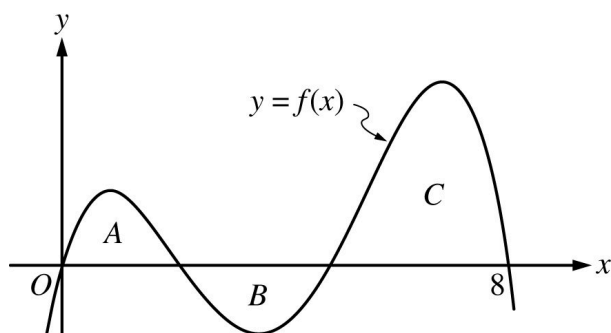
$$\frac{1}{c+5} \int_{-5}^c f(x) dx = \frac{1}{2}$$

$$\frac{1}{c+5}(-\pi + (-1) + 2 + c) = \frac{1}{2}$$

$$c = 3 + 2\pi$$

8-1

23.



The regions A , B , and C in the figure above are bounded by the graph of the function f and the x -axis. The area of region A is 14, the area of region B is 16, and the area of region C is 50. What is the average value of f on the interval $[0, 8]$?

(A) 6



(B) 10

(C) $\frac{40}{3}$ (D) $\frac{80}{3}$

(E) 48

8-1

24.

t (minutes)	0	4	8	12	16
$H(t)$ ($^{\circ}\text{C}$)	65	68	73	80	91

The temperature, in degrees Celsius ($^{\circ}\text{C}$), of an oven being heated is modeled by an increasing differentiable function H of time t , where t is measured in minutes. The table above gives the temperature as recorded every 4 minutes over a 16-minute period.

Write an integral expression in terms of H for the average temperature of the oven between time $t = 0$ and time $t = 16$. Estimate the average temperature of the oven using a left Riemann sum with four subintervals of equal length. Show the computations that lead to your answer.

Part B

1 point is earned for $\frac{1}{16} \int_0^{16} H(t) dt$

1 point is earned for left Riemann sum

1 point is earned for the answer

Average temperature is $\frac{1}{16} \int_0^{16} H(t) dt$

$$\frac{1}{16} \int_0^{16} H(t) dt = 4 \cdot (65 + 68 + 73 + 80)$$

$$\text{Average temperature} = \frac{4 \cdot 286}{16} = 71.5^{\circ}\text{C}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{1}{16} \int_0^{16} H(t) dt$

1 point is earned for left Riemann sum

1 point is earned for the answer

8-1

Average temperature is $\frac{1}{16} \int_0^{16} H(t) dt$

$$\frac{1}{16} \int_0^{16} H(t) dt = 4 \cdot (65 + 68 + 73 + 80)$$

$$\text{Average temperature} = \frac{4 \cdot 286}{16} = 71.5^\circ C$$

8-1

25. A tank contains 30 liters of water at time $t = 0$. For time $0 \leq t \leq 20$ minutes, water flows into the tank at a rate modeled by the function f defined by $f(t) = 2 \ln(t + 3) - 1.6$. For time $20 < t \leq 45$ minutes, water flows into the tank at a rate modeled by the function g defined by $g(t) = \frac{7.5}{\ln t} + 2.2$. Both $f(t)$ and $g(t)$ are measured in liters per minute.

**Part A**

Find the rate of change of g at time $t = 23$ minutes. Show the setup for your calculations, and indicate units of measure.

Part B

How many liters of water are in the tank at time $t = 20$ minutes? Show the setup for your calculations.

Part C

Let w be the function defined by $w(t) = \begin{cases} f(t) & \text{for } 0 \leq t \leq 20 \\ g(t) & \text{for } 20 < t \leq 45 \end{cases}$. Find $\lim_{t \rightarrow 20^+} w(t)$. Is w continuous at $t = 20$? Use the definition of continuity to justify your answer.

Part D

Consider the function w defined in part C. Find the average value of w from time $t = 15$ minutes to time $t = 30$ minutes. Show the setup for your calculations.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P1** is earned for a correct value either rounded or truncated to three places after the decimal and for labeling the value as $g'(23)$. An inappropriately rounded answer does not earn the point.
- **P1** can also be earned for the exact value $\frac{-7.5}{23(\ln 23)^2}$.

Model Solution:

$$g'(23) = -0.033$$



0	1
---	---

The response accurately includes the criteria below.

- $g'(23)$

8-1

Part A Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P2** is earned for correct units, whether or not they are attached to a numerical value for the rate of change.
- **P2** is also earned for the units “liters/minute².”

Model Solution:

The rate of change in the rate at time $t = 23$ is -0.033 liters per minute per minute.



0	1
---	---

The response accurately includes the criteria below.

- Units

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P3** is earned with a definite integral containing the correct integrand and correct lower and upper limits, with or without the differential.
- If the differential dt is missing:
 - Writing $\int_0^{20} f(t)$ or $30 + \int_0^{20} f(t)$ earns **P3**.
 - Because $\int_0^{20} f(t) + 30$ may mean $\int_0^{20} f(t)dt + 30$ or $\int_0^{20} (f(t) + 30)dt$:
 - Writing $\int_0^{20} f(t) + 30$ in the presence of a correct answer of 95.641 earns **P3**.
 - Writing $\int_0^{20} f(t) + 30$ in the presence of an answer of 95, 96, 95.6, or 95.64 earns **P3**.
 - Writing $\int_0^{20} f(t) + 30$ in the presence of no answer or an incorrect answer does not earn **P3**.
- Writing $\int f(t)dt$, $30 + \int f(t)dt$, $30 + \int f(t)$, or $\int f(t) + 30$ and never providing the definite integral, does not earn **P3**.

Model Solution:

8-1

$$30 + \int_0^{20} f(t)dt = 30 + \int_0^{20} (2 \ln(t+3) - 1.6)dt$$



0	1
---	---

The response accurately includes the criteria below.

- Definite integral

Part B Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- P4** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.
- Incorrect or unclear communication between the correct integral and the correct answer is treated as scratch work and is not considered in scoring. The following examples earn **P4** for the correct answer:
 - $\int_0^{20} f(t)dt = 65.641060$ so the number of liters of water is 95.641.
 - $\int_0^{20} f(t)dt = 65.641060 + 30 = 95.641060$ (Incorrect linkage is not considered in scoring.)
 - $\int_0^{20} f(t)dt = 95.641060$ (Incorrect linkage is not considered in scoring.)

Model Solution:

There are 95.641 liters of water in the tank at time $t = 20$ minutes.



0	1
---	---

The response accurately includes the criteria below.

- Answer

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

8-1

- To earn **P5**, a response must indicate $\lim_{t \rightarrow 20^+} w(t) = 4.704$ (or 4.703), either symbolically or verbally. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn **P5**, unless an earlier point was not earned due to inappropriate rounding.
 - A response of “the right-hand limit is 4.704” is sufficient to earn **P5**.

Model Solution:

$$\lim_{t \rightarrow 20^+} w(t) = \lim_{t \rightarrow 20^+} g(t) = 4.703562$$

The limit as t approaches 20 from the right is 4.704 (or 4.703).



0	1
---	---

The response accurately includes the criteria below.

- Right-hand limit

Part C Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P6**, a response must include $w(20) = 4.671$ (or 4.67 or 4.6), $\lim_{t \rightarrow 20^+} w(t) \neq w(20)$, and a conclusion of not continuous (with or without indicating $t = 20$).
- Alternate solution:

$$\lim_{t \rightarrow 20^-} w(t) = \lim_{t \rightarrow 20^-} f(t) = 4.670988$$

Because $\lim_{t \rightarrow 20^-} w(t) \neq \lim_{t \rightarrow 20^+} w(t)$, w is not continuous at $t = 20$.

Such a response is scored analogously to the model solution and earns **P6**.

- Special case: A response of “ $\lim_{t \rightarrow 20^+} w(t) = 4.7$ and $\lim_{t \rightarrow 20^-} w(t) = w(20) = 4.7$, therefore continuous” earns **P6**.

Model Solution:

$$w(20) = f(20) = 4.670988$$

Because $\lim_{t \rightarrow 20^+} w(t) \neq w(20)$, w is not continuous at $t = 20$.

8-1



0	1
---	---

The response accurately includes the criteria below.

- Answer with justification

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P7**, a response must present one or more definite integrals with correct integrands, with or without differentials. Limits of integration are not considered in scoring for **P7**. That is, any of the following earns **P7**: $\int_a^c w(t)dt$, $\int_a^b w(t)dt + \int_b^c w(t)dt$, or $\int_a^b f(t)dt + \int_b^c g(t)dt$, where a , b , and c are real numbers between 0 and 45, inclusive.

Model Solution:

$$\frac{1}{30-15} \int_{15}^{30} w(t)dt$$



0	1
---	---

The response accurately includes the criteria below.

- Definite integral

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P8**, a response must present a correct definite integral and evidence of division by 15.
 - A response that earned **P7** earns **P8** for $\frac{1}{15} \int_{15}^{30} w(t)dt$, $\frac{1}{15} \left(\int_{15}^{20} w(t)dt + \int_{20}^{30} w(t)dt \right)$, $\frac{1}{15} \left(\int_{15}^{20} f(t)dt + \int_{20}^{30} g(t)dt \right)$, or equivalent.

8-1

- Presenting the correct integral and the correct answer will suffice as evidence of division by 15 and will earn **P8**. Work presented between earning **P7** and the correct answer that does not reveal incorrect work done in the calculator will be treated as scratch work and will not be considered in scoring.
 - “ $\int_{15}^{30} w(t)dt = 67.559$, so 4.504” earns **P8** for correct limits of integration and an implied division by 15 made evident by the correct answer.
 - “ $\int_{15}^{30} w(t)dt = \frac{67.559}{15} = 4.504$ ” earns **P8** for correct limits of integration and an implied division by 15 made evident by the correct answer. The incorrect communication between the integral and the answer does not reveal incorrect work done in the calculator. This work is treated as scratch work and is not considered in scoring of **P8**.
- A response that did not earn **P7** is eligible to earn **P8** for $\frac{1}{15} \int_{15}^{30} f(t)dt$ or $\frac{1}{15} \int_{15}^{30} g(t)dt$.

Model Solution:

$$\begin{aligned} & \frac{1}{15} \left(\int_{15}^{20} w(t)dt + \int_{20}^{30} w(t)dt \right) \\ &= \frac{1}{15} \left(\int_{15}^{20} f(t)dt + \int_{20}^{30} g(t)dt \right) \end{aligned}$$



0	1
---	---

The response accurately includes the criteria below.

- Average value formula

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P9** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.

Model Solution:

The average value of w from time $t = 15$ and time $t = 30$ minutes is 4.504 (or 4.503) liters per minute.

8-1



0

1

The response accurately includes the criteria below.

- Answer

8-1

26.  A store is having a 12-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function S defined by $S(t) = 0.5t^4 - 16t^3 + 144t^2$ for $0 \leq t \leq 12$. At time $t = 0$, when the sale begins, there are no shoppers in the store.

- (a) At what rate are shoppers entering the store 3 hours after the start of the sale?
- (b) Find the value of $\frac{1}{3} \int_6^9 S'(t) dt$. Using correct units, explain the meaning of $\frac{1}{3} \int_6^9 S'(t) dt$ in the context of this problem.
- (c) The rate at which shoppers leave the store, measured in shoppers per hour, is modeled by the function L defined by $L(t) = -80 + \frac{4400}{t^2 - 14t + 55}$ for $0 \leq t \leq 12$. According to the model, how many shoppers are in the store at the end of the sale (time $t = 12$)? Give your answer to the nearest whole number.
- (d) Using the given models, find the time t , $0 \leq t \leq 12$, at which the number of shoppers in the store is the greatest. Justify your answer.

Part A

The response can earn up to 1 point:

1 point is earned for the answer

$$S'(3) = 486 \text{ shoppers/hour}$$



0	1
---	---

The student earns all of the following point:

1 point is earned for the answer

$$S'(3) = 486 \text{ shoppers/hour}$$

Part B

The response can earn up to 2 points:

1 point is earned for the value of $\frac{1}{3} \int_6^9 S'(t) dt$

$$\begin{aligned} \frac{1}{3} \int_6^9 S'(t) dt &= \frac{1}{3}(S(9) - S(6)) \\ &= \frac{1}{3}(3280.5 - 2376) = 301.5 \end{aligned}$$

1 point is earned for the meaning of $\frac{1}{3} \int_6^9 S'(t) dt$

8-1

$\frac{1}{3} \int_6^9 S'(t) dt$ is the average rate at which shoppers are entering the store in shoppers/hr between times $t = 6$ and $t = 9$ hours.



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for the value of $\frac{1}{3} \int_6^9 S'(t) dt$

$$\begin{aligned} \frac{1}{3} \int_6^9 S'(t) dt &= \frac{1}{3}(S(9) - S(6)) \\ &= \frac{1}{3}(3280.5 - 2376) = 301.5 \end{aligned}$$

1 point is earned for the meaning of $\frac{1}{3} \int_6^9 S'(t) dt$

$\frac{1}{3} \int_6^9 S'(t) dt$ is the average rate at which shoppers are entering the store in shoppers/hr between times $t = 6$ and $t = 9$ hours.

Part C

The response can earn up to 3 points:

1 point is earned for the integral

1 point is for use of $S(12)$

1 point is for the answer

$$S(12) - \int_0^{12} L(t) dt = 195.701684$$

Therefore, there are approximately 196 shoppers in the store at the end of the sale.



0	1	2	3
---	---	---	---

The student earns all of the following points:

1 point is earned for the integral

8-1

1 point is for use of $S(12)$

1 point is for the answer

$$S(12) - \int_0^{12} L(t) dt = 195.701684$$

Therefore, there are approximately 196 shoppers in the store at the end of the sale.

Part D

The response can earn up to 3 points:

1 point is earned for considering $S'(t) - L(t) = 0$

1 point is earned for the answer

1 point is earned for justification

The number of shoppers in the store at time t is given by the function N defined by

$$N(t) = S(t) - \int_0^t L(x) dx$$

$$N'(t) = S'(t) - L(t) = 0 \text{ when } t = 5.545066$$

t	$N(t)$
0	0
5.545066	1361.832842
12	195.702

The number of shoppers in the store is greatest at time $t = 5.545$ hours.



0	1	2	3
---	---	---	---

The student earns all of the following points:

1 point is earned for considering $S'(t) - L(t) = 0$

1 point is earned for the answer

1 point is earned for justification

The number of shoppers in the store at time t is given by the function N defined by

8-1

$$N(t) = S(t) - \int_0^t L(x) dx$$

$$N'(t) = S'(t) - L(t) = 0 \text{ when } t = 5.545066$$

t	$N(t)$
0	0
5.545066	1361.832842
12	195.702

The number of shoppers in the store is greatest at time $t = 5.545$ hours.

27. Tara's heart rate during a workout is modeled by the differentiable function h , where $h(t)$ is measured in beats per minute and t is measured in minutes from the start of the workout.

Which of the following expressions gives Tara's average heart rate from $t = 30$ to $t = 60$?

(A) $\int_{30}^{60} h(t) dt$

(B) $\frac{1}{30} \int_{30}^{60} h(t) dt$

(C) $\frac{1}{30} \int_{30}^{60} h'(t) dt$

(D) $\frac{h'(30) + h'(60)}{2}$

Answer B

Correct. The average value of a function f over an interval $[a, b]$ is $\frac{1}{b-a} \int_a^b f(x) dx$. Tara's average heart rate from $t = 30$ to $t = 60$ is the average value of the function h over the interval $[30, 60]$ and would therefore be given by the expression $\frac{1}{60-30} \int_{30}^{60} h(t) dt$.

8-1

28. Let g be a twice-differentiable function with $g'(x) > 0$ and $g''(x) > 0$ for all real numbers x , such that $g(3) = 12$ and $g(5) = 18$. Which of 20, 21, and 22 are possible values for $g(6)$?
- (A) 21 only
- (B) 22 only
- (C) 20 and 21 only
- (D) 21 and 22 only

Answer B

Correct. The graph of g is increasing because $g'(x) > 0$ and concave up because $g''(x) > 0$. The secant line through the points $(3, 12)$ and $(5, 18)$ is $y = 3(x - 3) + 12$. Because the graph of g is increasing and concave up, the graph will lie above the secant line for $x > 5$. In particular, the value of $g(6)$ is strictly greater than the value of y on the secant line at $x = 6$, that is, $g(6) > 3(6 - 3) + 12 = 21$. Therefore, 22 is the only possible value for $g(6)$.

8-1

29. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	60	90	120	135	150
$f(t)$ (gallons per second)	0	0.1	0.15	0.1	0.05	0

A customer at a gas station is pumping gasoline into a gas tank. The rate of flow of gasoline is modeled by a differentiable function f , where $f(t)$ is measured in gallons per second and t is measured in seconds since pumping began. Selected values of $f(t)$ are given in the table.

(a) Using correct units, interpret the meaning of $\int_{60}^{135} f(t) \, dt$ in the context of the problem. Use a right Riemann sum with the three subintervals $[60, 90]$, $[90, 120]$, and $[120, 135]$ to approximate the value of $\int_{60}^{135} f(t) \, dt$.

(b) Must there exist a value of c , for $60 < c < 120$, such that $f'(c) = 0$? Justify your answer.

(c) The rate of flow of gasoline, in gallons per second, can also be modeled by

$g(t) = \left(\frac{t}{500}\right) \cos\left(\left(\frac{t}{120}\right)^2\right)$ for $0 \leq t \leq 150$. Using this model, find the average rate of flow of gasoline over the time interval $0 \leq t \leq 150$. Show the setup for your calculations.

(d) Using the model g defined in part (c), find the value of $g'(140)$. Interpret the meaning of your answer in the context of the problem.

Part A

8-1

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the response must reference gallons of gasoline added/pumped and the time interval $t = 60$ to $t = 135$.

To earn the second point at least five of the six factors in the Riemann sum must be correct.

If there is any error in the Riemann sum, the response does not earn the third point.

A response of $(0.15)(30) + (0.1)(30) + (0.05)(15)$ earns both the second and third points, unless there is a subsequent error in simplification, in which case the response would earn only the second point.

A response that presents a correct value with accompanying work that shows the three products in the Riemann sum but does not show all six of the factors and/or the sum process, does not earn the second point but does earn the third point. For example, responses of either $4.5 + 3.0 + 0.75$ or $(0.15)(30), 0.1(30), 0.05(15) \rightarrow 8.25$ earn the third point but not the second.

A response of $f(90)(90 - 60) + f(120)(120 - 90) + f(135)(135 - 120) = 8.25$ earns both the second and the third points.

A response that presents an answer of only 8.25 does not earn either the second or third point.

A response that provides a completely correct left Riemann sum with accompanying work, $f(60)(30) + f(90)(30) + f(120)(15) = 9$, or $(0.1)(30) + (0.15)(30) + (0.1)(15)$ earns 1 of the last 2 points. A response with any errors or missing factors in a left Riemann sum earns neither of the last 2 points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Interpretation with units
- ☐ Form of Riemann sum
- ☐ Answer

Solution:

$\int_{60}^{135} f(t) dt$ represents the total number of gallons of gasoline pumped into the gas tank from time $t = 60$ seconds to time $t = 135$ seconds.

8-1

$$\int_{60}^{135} f(t) dt$$

$$\approx f(90)(90 - 60) + f(120)(120 - 90) + f(135)(135 - 120)$$

$$= (0.15)(30) + (0.1)(30) + (0.05)(15) = 8.25$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must present either $f(120) - f(60) = 0$, $0.1 - 0.1 = 0$ (perhaps as the numerator of a quotient), or $f(60) = f(120)$.

To earn the second point a response must:

- have earned the first point,
- state that f is continuous because f is differentiable (or equivalent), and
- answer “yes” in some way.

A response may reference either the Mean Value Theorem or Rolle’s Theorem.

A response that references the Intermediate Value Theorem cannot earn the second point.

✓

0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $f(120) - f(60) = 0$
- ☐ Answer with justification

Solution:

f is differentiable. $\Rightarrow f$ is continuous on $[60, 120]$.

$$\frac{f(120) - f(60)}{120 - 60} = \frac{0.1 - 0.1}{60} = 0$$

By the Mean Value Theorem, there must exist a c , for $60 < c < 120$, such that $f'(c) = 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

8-1

The exact value of $\frac{1}{150} \int_0^{150} g(t) dt$ is $\frac{12}{125} \sin(\frac{25}{16})$.

A response may present the average value formula in single or multiple steps. For example, the following response earns both points: $\int_0^{150} g(t) dt = 14.399504$ so the average rate is 0.0959967.

A response that presents the average value formula in multiple steps but provides incorrect or incomplete communication (e.g., $\int_0^{150} g(t) dt = \frac{14.399504}{150} = 0.0959967$) earns 1 out of 2 points.

A response of $\int_0^{150} g(t) dt = 0.0959967$ does not earn either point.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is generally eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $\frac{1}{150} \int_0^{150} g(t) dt = 0.149981$ or 0.002618.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Average value formula
- ☐ Answer

Solution:

$$\frac{1}{150-0} \int_0^{150} g(t) dt = 0.0959967$$

The average rate of flow of gasoline, in gallons per second, is 0.096 (or 0.095).

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The exact value of $g'(140)$ is $\frac{1}{500} \cos(\frac{49}{36}) - \frac{49}{9000} \sin(\frac{49}{36})$.

The value of $g'(140)$ may appear only in the interpretation.

To be eligible for the second point a response must present some numerical value for $g'(140)$.

8-1

To earn the second point the interpretation must include “the rate of flow of gasoline is changing at a rate of [the declared value of $g'(140)]$ ” and “at $t = 140$ ” (or equivalent).

An interpretation of “decreasing at a rate of -0.005 ” or “increasing at a rate of ” does not earn the second point.

Degree mode: In degree mode, $g'(140) = 0.001997$ or 0.00187 .



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $g'(140)$
- ☐ Interpretation

Solution:

$$g'(140) \approx -0.004908$$

$$g'(140) = -0.005 \text{ (or } -0.004)$$

The rate at which gasoline is flowing into the tank is decreasing at a rate of 0.005 (or 0.004) gallon per second per second at time $t = 140$ seconds.

8-1

30. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (years)	0	4	9	12
$R(t)$ (bears per year)	150	156	159	161

The rate at which the number of bears in a region is changing is modeled by the differentiable function R , where $R(t)$ is measured in bears per year and t is measured in years. Selected values of $R(t)$ are shown in the table.

(a) Approximate $R'(7)$ using the average rate of change of R over the interval $4 \leq t \leq 9$. Show the computations that lead to your answer, and indicate units of measure.

(b) Interpret the meaning of $\frac{1}{12} \int_0^{12} R(t) \, dt$ in the context of the problem. Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\frac{1}{12} \int_0^{12} R(t) \, dt$.

(c) The rate at which the number of bears in the region is changing can also be modeled by the function P defined by $P(t) = 150e^{0.02\sqrt{t}}$, where $P(t)$ is measured in bears per year and t is measured in years. There are 300 bears in the region at time $t = 0$. Use this model to find the number of bears in the region at time $t = 12$. Show the setup for your calculations, and give your answer to the nearest whole number of bears.

(d) Using the model defined in part (c), find the time t , for $0 \leq t \leq 12$, at which the number of bears in the region is a maximum. Justify your answer.

Part A

8-1

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must provide a numerical difference and a quotient. A response of only or earns the first point.

To earn the second point, a response must have “bear per year per year” (or the equivalent) attached to a numerical value.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation with supporting work
- ☐ Units

Solution:

$$R'(7) \approx \frac{R(9) - R(4)}{9 - 4} = \frac{159 - 156}{5} = 0.6$$

$$R'(7) \approx 0.6 \text{ bear per year per year}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, a response must provide “average,” “rate of change,” and “from time $t = 0$ to $t = 12$ ” (or equivalent). Units are not required to earn the first point.

To earn the second point, at least five of the six factors in the Riemann sum must be correct. The factor of $\frac{1}{12}$ is assessed in the third point.

If there is any error in the Riemann sum, the response does not earn the third point.

A response of only $\frac{1}{12}(150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3)$ earns the second and third points.

A response that presents the correct average value with accompanying work that does not explicitly present the six factors and/or the sum process does not earn the second point but will earn the third point. For example, responses of “ $\frac{150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3}{12}$ ” or “600, 780, 477 $\rightarrow \frac{1857}{12}$ ” would earn the third point but not the second.

A response that presents an answer of $\frac{156 \cdot 4 + 159 \cdot 5 + 161 \cdot 3}{12}$ (which uses a right Riemann sum) earns 1 of the last 2 points. If this response is simplified, the result must be $\frac{1902}{12}$, $\frac{951}{6}$, $\frac{317}{2}$, or 158.5.

8-1

An unsupported response of $\frac{1857}{12}$, $\frac{619}{4}$, or 154.75 does not earn the second or third point.

A response that miscommunicates by equating two unequal quantities, e.g.,

$$\int_0^{12} R(t) \, dt = \frac{150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3}{12}, \text{ does not earn the third point.}$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Interpretation
- ☐ Form of left Riemann sum
- ☐ Answer

Solution:

$\frac{1}{12} \int_0^{12} R(t) \, dt$ is the average rate of change in the number of bears in the region, in bears per year, from time $t = 0$ to time $t = 12$ years.

$$\begin{aligned} \int_0^{12} R(t) \, dt &\approx R(0) \cdot (4 - 0) + R(4) \cdot (9 - 4) + R(9) \cdot (12 - 9) \\ &= 150 \cdot (4 - 0) + 159 \cdot (9 - 4) + 159 \cdot (12 - 9) \\ &= 150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3 = 1857 \end{aligned}$$

$$\frac{1}{12} \int_0^{12} R(t) \, dt \approx \frac{1857}{12} = \frac{619}{4} = 154.75 \text{ bears per year}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a definite integral $\int_a^b P(t) \, dt$ with any limits of integration. If the limits are incorrect, the response does not earn the second point.

The second point cannot be earned without the first point, and is earned only for an answer of 2185 or 2185.339 or 2185.338.

Missing differentials:

- A response of only $\int_0^{12} P(t)$ or $\int_0^{12} P(t) = 1885.3389$ earns the first point but not the second.

8-1

- A response of $300 + \int_0^{12} P(t)$ earns the first point and in the presence of the correct answer earns the second point.
- A response of $\int_0^{12} P(t) + 300$ does not earn the first point but earns the second point for the correct answer.

A response that miscommunicates, e.g., $\int_0^{12} P(t) dt = 1885.3389 + 300$, does not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Integral
- ☐ Answer

Solution:

$$300 + \int_0^{12} P(t) dt = 300 + 1885.3389 = 2185.3389$$

At $t = 12$ years, there are 2185 bears in the region.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that considers $P = 0$ earns the first point.

A response of “at time $t = 12$ ” with no justification does not earn the second point.

The second point cannot be earned for any answer other than time $t = 12$.

Alternate solution using Candidates Test:

$P(t) = 0$ has no solutions in the interval $0 \leq t \leq 12$, so the maximum number of bears occurs at an endpoint of the interval.

The region has 300 bears at time $t = 0$.

From part (c), the region has 2185 bears at time $t = 12$.

For $0 \leq t \leq 12$, the maximum number of bears is at time $t = 12$ years.

8-1

- The scoring is the same as the model solution.
- A response using this argument may import an incorrect value for the number of bears in the region in year 12 from part (c), provided that value is larger than 300.
- A response using this argument need not report the number of bears at time $t = 12$. Stating that this number is greater than 300 is sufficient.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers sign of P
- ☐ Answer with justification

Solution:

$P(t) > 0$ for all $t \geq 0$, so the bear population is always increasing.

For $0 \leq t \leq 12$, the maximum number of bears is at time $t = 12$ years.

8-1

31. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (years)	0	10	22	30	40
$A(t)$ (hundred pounds per year)	163	142	124	118	113

The amount of nitrogen in a particular river is affected by two sources. Nitrogen from the first source enters the river at a rate modeled by the decreasing, differentiable function A . Selected values of $A(t)$ are shown in the table above. Nitrogen from the second source enters the river at a rate modeled by the function B defined by $B(t) = 20 + \left(\frac{95}{1+\sqrt[3]{t}}\right)^{1.25}$. Both $A(t)$ and $B(t)$ are measured in hundreds of pounds of nitrogen per year, and t is measured in years for $0 \leq t \leq 40$.

(a) Using a right Riemann sum with the four subintervals indicated by the data in the table, approximate the total amount of nitrogen from the first source that enters the river over the time interval $0 \leq t \leq 40$. Is this approximation an overestimate or an underestimate? Give a reason for your answer.

(b) Find $B'(39)$. Using correct units, interpret the meaning of $B'(39)$ in the context of the problem.

(c) Based on model B , find the average rate, in hundreds of pounds per year, at which nitrogen from the second source enters the river during the time interval $5 \leq t \leq 35$.

(d) Using model B and your approximation from part (a), find an approximation for the total combined amount of nitrogen from the two sources that enters the river over the time interval $0 \leq t \leq 40$. Give your answer to the nearest hundred pounds.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

8-1

The first point is earned for a sum of four products with at most one error among the 8 factors. If there is an error in the setup, the second point is not earned.

A response of $10 \cdot A(10) + 12 \cdot A(22) + 8 \cdot A(30) + 10 \cdot A(40)$ would earn the first point, but not yet the second. The first two points are earned for $10 \cdot 142 + 12 \cdot 124 + 8 \cdot 118 + 10 \cdot 113$.

Without the products, the sum $1420 + 1488 + 944 + 1130$ by itself would earn the second point but not the first point.

Once the first point is earned it cannot be lost by incorrectly using values from the table or performing some type of operation on the Riemann sum.

A completely correct setup and approximation using a left Riemann sum $(10 \cdot 163 + 12 \cdot 142 + 8 \cdot 124 + 10 \cdot 118 = 5506)$ will earn one of the first two points and is not eligible for the third point. Any error in the setup or evaluation using a left Riemann sum will earn no points.

Units are not scored in this part, except an approximation of 498,200 must clearly be labeled as *pounds* to earn the second point.

We will interpret the word “function” to mean A , unless otherwise prescribed.

The response must indicate which function is decreasing. A response of “underestimate because it (or the graph, or the curve) is decreasing” will not earn the third point.

The third point will not be earned if the response concludes the approximation is an underestimate because the table values are decreasing.

The third point will not be earned if incorrect statements are made in addition to saying A is decreasing.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Right Riemann sum setup
- ☐ Approximation
- ☐ Underestimate with reason

Solution:

$$\begin{aligned}
 \int_0^{40} A(t) \, dt &\approx (10 - 0) \cdot A(10) + (22 - 10) \cdot A(22) \\
 &+ (30 - 22) \cdot A(30) + (40 - 30) \cdot A(40) \\
 &= 10 \cdot 142 + 12 \cdot 124 + 8 \cdot 118 + 10 \cdot 113 = 4982
 \end{aligned}$$

8-1

Approximately 4982 hundred pounds of nitrogen from the first source enters the river over the time interval $0 \leq t \leq 40$.

The approximation is an underestimate because a right Riemann sum is used and A is decreasing.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The value of $B'(39)$ may appear in the interpretation. No work is required in order to earn the first point.

In the presence of a correct value for $B'(39)$, any symbolic attempt at differentiation will be ignored.

Any unevaluated expression equivalent to $1.25 \left(\frac{95}{1 + \sqrt[3]{39}} \right)^{0.25} \left(\frac{-95}{\left(1 + \sqrt[3]{39}\right)^2} \right) \left(\frac{1}{3}(39)^{-2/3} \right)$ will earn the first point.

The second point cannot be earned if no value for $B'(39)$ is presented.

The units (hundred pounds per year per year) may be attached to the value of $B'(39)$ and if so, they do not need to be repeated in the interpretation.

If units attached to $B'(39)$ do not agree with units in the interpretation, read the units in the interpretation.

The interpretation must include time $t = 39$, rate of change of the rate at which nitrogen enters the river (or that the rate that nitrogen enters the river is decreasing at the stated rate), and units of hundred pounds per year per year.

The second point cannot be earned with an incorrect statement such as "... is decreasing at a rate of $-0.385 \dots$ " or the statement "... is increasing at a rate of $-0.385 \dots$ ".

Use of the term "acceleration" of the amount of nitrogen coming from the second source will not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Value
- ☐ Interpretation

Solution:

$$B'(39) = -0.384963$$

8-1

At time $t = 39$ years, the rate at which nitrogen from the second source enters the river was decreasing at a rate of 0.385 (or 0.384) hundred pounds per year per year.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for presenting the definite integral $\int_5^{35} B(t) dt$.

Dividing by 30 will be scored in the second point.

Once the first point is earned, it cannot be lost.

If the first point is not earned because of a copy error in the integrand, the response can earn the second point for the correct answer.

An unsupported answer of 80.063 does not earn the second point.

Units are not scored in this part, except an answer of 8006.3 must clearly be labeled as *pounds per year* in order to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Definite integral
- ☐ Answer

Solution:

$$\frac{1}{35-5} \int_5^{35} B(t) dt = 80.063385$$

During the time interval $5 \leq t \leq 35$, the average rate at which nitrogen from the second source enters the river was 80.063 hundred pounds per year.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

For any $c > 0$, the presence of $c \int_0^{40} B(t) dt$ will earn the first point.

8-1

A copy error in the integrand imported from part (c) will earn the first point. Any new copy error will not earn the first point. In either case the second point can be earned if the correct answer is given.

Read for correctness or consistency with any positive value of $\int_0^{40} A(t) dt$ imported from part (a).

The second point is not earned for a correct value with no supporting work.

The second point is earned for an answer that is one of the four values 8368, 8369, 8368.35, or 8368.351 (with supporting work).

If one of the values 8368, 8369, 8368.35, or 8368.351 is presented but then rounded to 8400 (e.g., $8368 \approx 8400$), the response will earn the second point. However, a response of just 8400, without one of these four values present, will not earn the second point.

Units are not scored in this part, except an answer of 836,800 (or 836,900) must clearly be labeled as *pounds* in order to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $\int_0^{40} B(t) dt$
- ☐ Answer

Solution:

$$\text{Total} = \int_0^{40} A(t) dt + \int_0^{40} B(t) dt \approx 4982 + 3386.350701 \approx 8368$$

The total combined amount of nitrogen from the two sources that enters the river over the time interval $0 \leq t \leq 40$ is approximately 8368 hundred pounds.

8-1

32. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Diego runs back and forth along a straight track. During the time interval $0 \leq t \leq 30$ seconds, Diego's velocity, in feet per second, is modeled by the function v given by $v(t) = \frac{200 \sin\left(\frac{t^2}{75}\right)}{t+1}$.

- What is the first time, t_1 , that Diego changes direction? Find Diego's average velocity over the time interval $0 \leq t \leq t_1$.
- At time $t = 30$, how far is Diego from his position at time $t = 0$?
- Find the total distance that Diego runs during the 30 seconds from time $t = 0$ to time $t = 30$.
- Let $a(t)$ represent Diego's acceleration during the time interval $0 < t < 30$ seconds. Using correct units, explain the meaning of $\int_5^{10} a(t) \, dt$ in the context of the problem. Find the value of $\int_5^{10} a(t) \, dt$.

Part A

For the first point, supporting work stating that $v(t)$ changes sign is required.

For the second point, supporting work that includes the average value expression with an appropriate definite integral is required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

8-1

- ☐ $t_1 = 15.350$
- ☐ average value integral
- ☐ answer

Solution:

Diego changes direction when $v(t)$ changes sign. The first time this happens is at $t_1 = 15.350$ (or 15.349).

$$\frac{1}{t_1 - 0} \int_0^{t_1} v(t) dt = 10.503965$$

Diego's average velocity over the time interval $0 \leq t \leq t_1$ is 10.504 (or 10.503) feet per second.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ definite integral
- ☐ answer

Solution:

$$\int_0^{30} v(t) dt = 128.322555$$

At time $t = 30$, Diego is 128.323 (or 128.322) feet from his position at time $t = 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

8-1

- ☐ definite integral
- ☐ answer

Solution:

$$\int_0^{30} |v(t)| dt = 243.425185$$

During the 30 seconds from time $t = 0$ to time $t = 30$, the total distance that Diego runs is 243.425 feet.

Part D

The first point requires 3 things: (1) net change in Diego's velocity, (2) units of feet per second, and (3) reference to time interval from $t = 5$ to $t = 10$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ interpretation with units
- ☐ answer

Solution:

Since acceleration is the rate of change of velocity, $\int_5^{10} a(t) dt$ is the net change in Diego's velocity, in feet per second, during the 5 seconds from time $t = 5$ to time $t = 10$.

$$\int_5^{10} a(t) dt = \int_5^{10} v'(t) dt = v(10) - v(5) = 6.765 \text{ feet per second}$$

8-1

33. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Emma and Lily jog in the same direction along a straight track. For $0 \leq t \leq 15$, Emma's velocity at time t is given by $E(t) = \frac{7510}{t^2 - 7t + 80.22}$ and Lily's velocity at time t is given by $L(t) = 12t^3e^{-0.5t}$. Both $E(t)$ and $L(t)$ are positive for $0 \leq t \leq 15$ and are measured in meters per minute, and t is measured in minutes. Emma is 10 meters ahead of Lily at time $t = 0$, and Emma remains ahead of Lily for $0 < t \leq 15$.

- Find the value of $\frac{1}{6} \int_2^8 E(t) \, dt$. Using correct units, interpret the meaning of $\frac{1}{6} \int_2^8 E(t) \, dt$ in the context of the problem.
- At time $t = 5$, is Emma speeding up or slowing down? Give a reason for your answer.
- Is the distance between Emma and Lily at time $t = 7$ increasing or decreasing? Give a reason for your answer.
- What is the maximum distance between Emma and Lily over the time interval $0 \leq t \leq 15$? Justify your answer.

Part A

For the second point, the response must include reference to average value over the correct time interval with correct units.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

8-1

The student response accurately includes both of the criteria below.

- ☐ value
- ☐ meaning

Solution:

$$\frac{1}{6} \int_2^8 E(t) dt = 103.178012$$

Emma's average velocity over the time interval $2 \leq t \leq 8$ is 103.178 meters per minute.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ uses $E(5)$ and $E'(5)$
- ☐ answer with reason

Solution:

$$E(5) > 0 \text{ and } E'(5) = -4.569193 < 0$$

At time $t = 5$, since Emma has positive velocity and negative acceleration, she is slowing down.

Part C

For the first point, $E(7) - L(7) < 0$ is required with or without numerical values.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct answer with reason.

Solution:

8-1

Emma's velocity at $t = 7$ is $E(7) = 93.617552$.

Lily's velocity at time $t = 7$ is $L(7) = 124.292430$.

Since $E(7) - L(7) < 0$, the distance between Emma and Lily at $t = 7$ is decreasing.

Part D

The third point is earned with correct use of the Candidates Test.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ considers $E(t) - L(t) = 0$
- ☐ $t = 4.233$ and $t = 11.671$ as candidates
- ☐ justification
- ☐ answer

Solution:

The distance between Emma and Lily at time t is given by $10 + \int_0^t (E(x) - L(x)) dx$.

$$\frac{d}{dt} \left(10 + \int_0^t (E(x) - L(x)) dx \right) = E(t) - L(t) = 0$$

$$\Rightarrow t_1 = 4.232614 \text{ and } t_2 = 11.671403$$

t	Distance between Emma and Lily
0	10
t_1	266.998916
t_2	126.880323
15	156.143676

The maximum distance between Emma and Lily is 267 meters.

8-1

34. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Gwen runs back and forth along a straight track. During the time interval $0 \leq t \leq 45$ seconds, Gwen's velocity, in feet per second, is modeled by the function v given by $v(t) = \frac{250 \sin\left(\frac{t^2}{120}\right)}{t+6}$.

- What is the first time, t_1 , that Gwen changes direction? Find Gwen's average velocity over the time interval $0 \leq t \leq t_1$.
- At time $t = 45$, how far is Gwen from her position at time $t = 0$?
- Find the total distance that Gwen runs during the 45 seconds from time $t = 0$ to time $t = 45$.
- Let $a(t)$ represent Gwen's acceleration during the time interval $0 < t < 45$ seconds. Find the value of $\int_{15}^{30} a(t) \, dt$. Using correct units, explain the meaning of $\int_{15}^{30} a(t) \, dt$ in the context of the problem.

Part A

For the first point, supporting work stating that $v(t)$ changes sign is required.

For the second point, supporting work that includes the average value expression with an appropriate definite integral is required. To be eligible for the second point, the presented value of t_1 may be incorrect but must be positive.

The third point requires the first two points to have been earned.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

8-1



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $t_1 = 19.416$
- ☐ average value integral
- ☐ answer

Solution:

Gwen changes direction when $v(t)$ changes sign. The first time this happens is at $t_1 = 19.416$.

$$\frac{1}{t_1 - 0} \int_0^{t_1} v(t) dt = 7.264377$$

Gwen's average velocity over the time interval $0 \leq t \leq t_1$ is 7.264 feet per second.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ definite integral
- ☐ answer

Solution:

$$\int_0^{45} v(t) dt = 116.537972$$

At time $t = 45$, Gwen is 116.538 (or 116.537) feet from her position at time $t = 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

8-1



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ definite integral
- ☐ answer

Solution:

$$\int_0^{45} |v(t)| dt = 250.099483$$

During the 45 seconds from time $t = 0$ to time $t = 45$, the total distance that Gwen runs is 250.099 feet.

Part D

The second point requires 3 things: (1) net change in Gwen's velocity, (2) units of feet per second, and (3) reference to time interval from $t = 15$ to $t = 30$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ interpretation with units

Solution:

$$\int_{15}^{30} a(t) dt = \int_{15}^{30} v'(t) dt = v(30) - v(15) = -4.844 \text{ feet per second}$$

Since acceleration is the rate of change of velocity, $\int_{15}^{30} a(t) dt$ is the net change in Gwen's velocity, in feet per second, during the 15 seconds from time $t = 15$ to time $t = 30$.

8-1

35. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Mary and Chance walk in the same direction along a straight path. For $0 \leq t \leq 20$, Mary's velocity at time t is given by $M(t) = \frac{6010}{t^2 - 3t + 50.5}$ and Chance's velocity at time t is given by $C(t) = 8.5t^3e^{-0.45t}$. Both $M(t)$ and $C(t)$ are positive for $0 \leq t \leq 20$ and are measured in meters per minute, and t is measured in minutes. Mary is 12 meters ahead of Chance at time $t = 0$, and Mary remains ahead of Chance for $0 < t \leq 20$.

- (a) Find the value of $\frac{1}{10} \int_5^{15} M(t) \, dt$. Using correct units, interpret the meaning of $\frac{1}{10} \int_5^{15} M(t) \, dt$ in the context of the problem.
- (b) At time $t = 10$, is Mary speeding up or slowing down? Give a reason for your answer.
- (c) Is the distance between Mary and Chance at time $t = 18$ increasing or decreasing? Give a reason for your answer.
- (d) What is the maximum distance between Mary and Chance over the time interval $0 \leq t \leq 20$? Justify your answer.

Part A

For the second point, the response must include reference to average value over the correct time interval with correct units.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

8-1

The student response accurately includes both of the criteria below.

- ☐ value
- ☐ meaning

Solution:

$$\frac{1}{10} \int_5^{15} M(t) dt = 54.409167$$

Mary's average velocity over the time interval $5 \leq t \leq 10$ is 54.409 meters per minute.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ uses $M(10)$ and $M'(10)$
- ☐ answer with reason

Solution:

$$M(10) > 0 \text{ and } M'(10) = -7.036380 < 0$$

At time $t = 10$, since Mary has positive velocity and negative acceleration, she is slowing down.

Part C

For the point, $M(18) - C(18) > 0$ is required with or without numerical values.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct answer with reason.

Solution:

8-1

Mary's velocity at time $t = 18$ is $M(18) = 18.751950$. Chance's velocity at time $t = 18$ is $C(18) = 15.047042$.

Since $M(18) - C(18) > 0$, the distance between Mary and Chance at time $t = 18$ is increasing.

Part D

The third point is earned with correct use of the Candidates Test.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ considers $M(t) - C(t) = 0$
- ☐ $t = 4.584$ and $t = 16.715$ as candidates
- ☐ justification
- ☐ answer

Solution:

The distance between Mary and Chance at time t is given by $12 + \int_0^t (M(x) - C(x)) dx$.

$$\frac{d}{dt} \left(12 + \int_0^t (M(x) - C(x)) dx \right) = M(t) - C(t) = 0$$

$$\Rightarrow t_1 = 4.583625 \text{ and } t_2 = 16.714620$$

t	Distance between Mary and Chance
0	12
t_1	365.527469
t_2	13.344315
20	27.010082

The maximum distance between Mary and Chance is 365.527 meters.

8-1

36.  The penguin population on an island is modeled by a differentiable function P of time t , where $P(t)$ is the number of penguins and t is measured in years, for $0 \leq t \leq 40$. There are 100,000 penguins on the island at time $t = 0$. The birth rate for the penguins on the island is modeled by

$$B(t) = 1000e^{0.06t} \text{ penguins per year}$$

and the death rate for the penguins on the island is modeled by

$$D(t) = 250e^{0.1t} \text{ penguins per year.}$$

- (a) What is the rate of change of the penguin population on the island at time $t = 0$?
- (b) To the nearest whole number, what is the penguin population on the island at time $t = 40$?
- (c) To the nearest whole number, what is the average rate of change of the penguin population on the island for $0 \leq t \leq 40$?
- (d) To the nearest whole number, find the absolute minimum penguin population and the absolute maximum penguin population on the island for $0 \leq t \leq 40$. Show the analysis that leads to your answer.

Part A

The response can earn up to 1 point:

1 point is earned for the answer

$$P'(0) = B(0) - D(0) = 1000 - 250 = 750 \text{ penguins per year}$$



0	1
---	---

The response earns all of the following point:

1 point is earned for the answer

$$P'(0) = B(0) - D(0) = 1000 - 250 = 750 \text{ penguins per year}$$

Part B

The response can earn up to 2 points:

1 point is earned for correct limits

1 point is earned for the integrand

1 point is earned for answer

$$P(40) = 100000 + \int_0^{40} (B(t) - D(t)) dt$$

$$= 100000 + 33057.56459$$

8-1

There are 133,058 penguins on the island.



0	1	2	3
---	---	---	---

The student earns all of the following point:

1 point is earned for correct limits

1 point is earned for the integrand

1 point is earned for answer

$$P(40) = 100000 + \int_0^{40} (B(t) - D(t)) dt$$

$$= 100000 + 33057.56459$$

There are 133,058 penguins on the island.

Part C

The response can earn up to 1 point:

1 point is earned for the answer

$$\frac{1}{40} \int_0^{40} (B(t) - D(t)) dt = 826.439$$

OR

$$\frac{P(40) - P(0)}{40 - 0} = \frac{133058 - 100000}{40} = 826.45$$

The average rate of change is 826 penguins per year.



0	1
---	---

The student earns all of the following point:

1 point is earned for the answer

$$\frac{1}{40} \int_0^{40} (B(t) - D(t)) dt = 826.439$$

8-1

OR

$$\frac{P(40) - P(0)}{40 - 0} = \frac{133058 - 100000}{40} = 826.45$$

The average rate of change is 826 penguins per year.

Part D

The response can earn up to 4 points:

1 point is earned for the expression $B(t) - D(t) = 0$

$$B(t) - D(t) = 0$$

1 point is earned for the solution for t

$$1000e^{0.06t} = 250e^{0.1t} \Rightarrow t = A = \frac{\ln 4}{0.04} = 34.657359$$

The absolute minimum and absolute maximum occur at a critical point or at an endpoint.

1 point is earned for the minimum value

1 point is earned for the maximum value

$$P(0) = 100000$$

$$P(A) = 100000 + \int_0^A (B(t) - D(t)) dt = 139166.667$$

$$P(40) = 133058$$

The minimum population is 100,000 and the maximum population is 139,167 penguins.



0	1	2	3	4
---	---	---	---	---

The student earns all of the following point:

1 point is earned for the expression $B(t) - D(t) = 0$

$$B(t) - D(t) = 0$$

1 point is earned for the solution for t

$$1000e^{0.06t} = 250e^{0.1t} \Rightarrow t = A = \frac{\ln 4}{0.04} = 34.657359$$

8-1

The absolute minimum and absolute maximum occur at a critical point or at an endpoint.

1 point is earned for the minimum value

1 point is earned for the maximum value

$$P(0) = 100000$$

$$P(A) = 100000 + \int_0^A (B(t) - D(t)) dt = 139166.667$$

$$P(40) = 133058$$

The minimum population is 100,000 and the maximum population is 139,167 penguins.

37.  On a certain day, an industrial plant manufactures steel at a rate modeled by $R(t) = 10 + 30e^{-0.6t}$ tons per hour, where t is measured in hours since the plant opened that day. What is the average rate at which the plant manufactures steel, in tons per hour, over the time interval $0 \leq t \leq 5$?

(A) 11.494

(B) 19.502

(C) 25.747

(D) 97.511

**Answer B**

Correct. The average rate at which the plant manufactures steel over the time interval $0 \leq t \leq 5$ is given by the average value of R over this interval. The average value of a function f on an interval $a \leq t \leq b$ is given by

$$\frac{1}{b-a} \int_a^b f(t) dt.$$
$$\frac{1}{5} \int_0^5 R(t) dt = \frac{1}{5} \int_0^5 (10 + 30e^{-0.6t}) dt = 19.052$$

The calculator is used to evaluate the definite integral.

8-1

38. If the average value of the function f on the interval $1 \leq x \leq 4$ is 8, what is the value of

$$\int_1^4 (3f(x) + 2x)dx ?$$

- (A) 30
(B) 39
(C) 78
(D) 87

**Answer D**

Correct. Since the average value of f on the interval $1 \leq x \leq 4$ is 8, $\frac{1}{4-1} \int_1^4 f(x)dx = 8 \Rightarrow \int_1^4 f(x)dx = 24$.

The rules of integration are used to compute the following.

$$\int_1^4 (3f(x) + 2x)dx = 3 \int_1^4 f(x)dx + \int_1^4 2x dx = 3(24) + x^2 \Big|_1^4 = 72 + (16 - 1) = 87$$

39.  Let f be the function given by $f(x) = \sin(2x) \cos(1+x)$. What is the average value of f on the closed interval $1 \leq x \leq 3$?

- (A) 0.739
(B) 0.369
(C) 0.281
(D) -0.098

**Answer B**

Correct. The average value of f over the interval $[1, 3]$ is computed with a definite integral as follows. The calculator is used to do the numerical integration.

$$\frac{1}{3-1} \int_1^3 f(x) dx = \frac{1}{2} \int_1^3 \sin(2x) \cos(1+x) dx = 0.369$$

8-1

40. The intensity of light at a distance x meters from a source is modeled by the function R given by $R(x) = \frac{k}{x^2}$, where k is a positive constant. Which of the following gives the average intensity of the light between 4 meters and 6 meters from the source?
- (A) $\frac{k}{25}$
- (B) $-\frac{k}{6} + \frac{k}{4}$
- (C) $\frac{1}{2} \left(-\frac{k}{6} + \frac{k}{4} \right)$ ✓
- (D) $\frac{1}{2} \left(\frac{k}{16} + \frac{k}{36} \right)$

Answer C

Correct. The average value of R over the interval $[4, 6]$ is computed with a definite integral.

$$\frac{1}{6-4} \int_4^6 R(x) dx = \frac{1}{2} \int_4^6 \frac{k}{x^2} dx = \frac{1}{2} \left(-\frac{k}{x} \Big|_4^6 \right) = \frac{1}{2} \left(-\frac{k}{6} + \frac{k}{4} \right)$$

41. The average value of a function f over the interval $[-2, 3]$ is -6 , and the average value of f over the interval $[3, 5]$ is 20 . What is the average value of f over the interval $[-2, 5]$?
- (A) 2
- (B) 7
- (C) $\frac{10}{7}$ ✓
- (D) 5

Answer C

Correct. The information about the average value of f over the intervals $[-2, 3]$ and $[3, 5]$ gives the following.

$$\frac{1}{5} \int_{-2}^3 f(x) dx = -6 \Rightarrow \int_{-2}^3 f(x) dx = -30$$

$$\frac{1}{2} \int_3^5 f(x) dx = 20 \Rightarrow \int_3^5 f(x) dx = 40$$

Therefore, the average value of f over the interval $[-2, 5]$ is

$$\frac{1}{5-(-2)} \int_{-2}^5 f(x) dx = \frac{1}{7} \left(\int_{-2}^3 f(x) dx + \int_3^5 f(x) dx \right) = \frac{1}{7} (-30 + 40) = \frac{10}{7}.$$

8-1

42. The intensity of radiation at a distance x meters from a source is modeled by the function R given by $R(x) = \frac{k}{x^2}$, where k is a positive constant. Which of the following gives the average intensity of radiation between 10 meters and 50 meters from the source?
- (A) $\frac{k}{900}$
- (B) $-\frac{k}{50} + \frac{k}{10}$
- (C) $\frac{1}{40} \left(-\frac{k}{50} + \frac{k}{10} \right)$ ✓
- (D) $\frac{1}{2} \left(\frac{k}{100} + \frac{k}{2500} \right)$

Answer C

Correct. The average value of $R(t)$ over the interval $[10, 50]$ is computed with a definite integral.

$$\frac{1}{50-10} \int_{10}^{50} R(x) dx = \frac{1}{40} \int_{10}^{50} \frac{k}{x^2} dx = \frac{1}{40} \left(-\frac{k}{x} \Big|_{10}^{50} \right) = \frac{1}{40} \left(-\frac{k}{50} + \frac{k}{10} \right)$$

43.  Let f be the function given by $f(x) = 3 \ln(2 + x^2) \cos x$. What is the average value of f on the closed interval $2 \leq x \leq 6$?
- (A) -6.691
- (B) -1.673 ✓
- (C) 3.179
- (D) 4.121

Answer B

Correct. The average value of f over the interval $[2, 6]$ is computed with a definite integral as follows. The calculator is used to do the numerical integration.

$$\frac{1}{6-2} \int_2^6 f(x) dx = \frac{1}{4} \int_2^6 3 \ln(2 + x^2) \cos x dx = -1.673$$

8-1

44. If the average value of the function f on the interval $2 \leq x \leq 6$ is 3, what is the value of

$$\int_2^6 (5f(x) + 2)dx ?$$

- (A) 17
(B) 23
(C) 62
(D) 68

**Answer D**

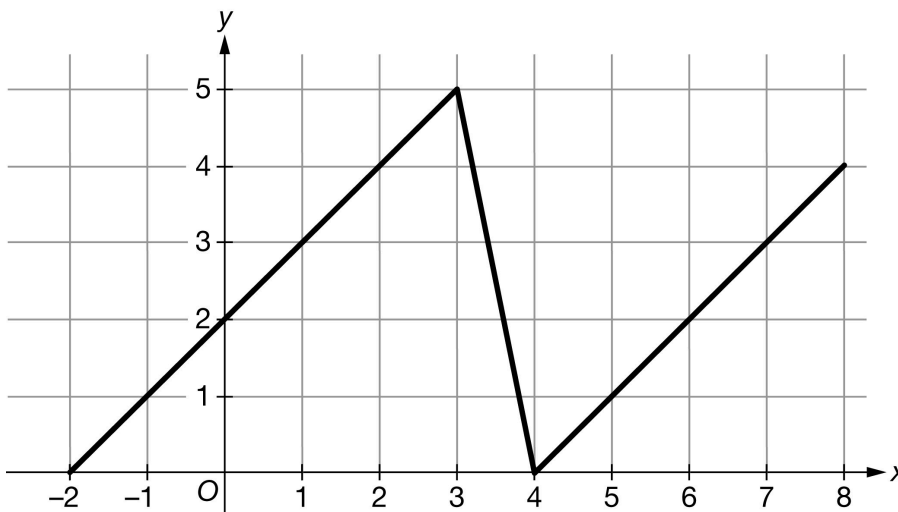
Correct. Since the average value of f on the interval $2 \leq x \leq 6$ is 3, $\frac{1}{4} \int_2^6 f(x)dx = 3 \Rightarrow \int_2^6 f(x)dx = 12$.

The rules of integration are used to compute the following.

$$\int_2^6 (5f(x) + 2)dx = \int_2^6 5f(x)dx + \int_2^6 2dx = 5 \int_2^6 f(x)dx + \int_2^6 2dx = 5(12) + 2(4) = 68$$

8-1

45.



The graph of the continuous function f consists of three line segments, as shown above. What is the average value of f on the interval $[-2, 8]$?

- (A) 0.4
(B) 2.0
(C) 2.3
(D) 2.5

**Answer C**

Correct. The average value of a continuous function f over the interval $a \leq x \leq b$ is $\frac{1}{b-a} \int_a^b f(x) dx$. Applying that formula to this problem gives the following calculation.

$$\frac{1}{8-(-2)} \int_{-2}^8 f(x) dx = \frac{1}{10} \left(\frac{1}{2} \cdot 6 \cdot 5 + \frac{1}{2} \cdot 4 \cdot 4 \right) = \frac{1}{10} (15 + 8) = 2.3$$

8-1

46.  Let f be the function given by $f(x) = 3x \sin x$. What is the average value of f on the closed interval $1 \leq x \leq 7$?
- (A) -14.764
- (B) -2.461 ✓
- (C) 1.879
- (D) 8.161

Answer B

Correct. The average value of a function f over the interval $a \leq x \leq b$ is $\frac{1}{b-a} \int_a^b f(x) dx$. Applying that formula to this problem gives the following calculation that can be done with the calculator.

$$\frac{1}{7-1} \int_1^7 3x \sin x \, dx = -2.461$$

47. The average value of a function f over the interval $[-1, 2]$ is -4 , and the average value of f over the interval $[2, 7]$ is 8 . What is the average value of f over the interval $[-1, 7]$?
- (A) $\frac{1}{2}$
- (B) 2
- (C) $\frac{7}{2}$ ✓
- (D) 14

Answer C

Correct. The information about the average value of f over the intervals $[-1, 2]$ and $[2, 7]$ gives the following.

$$\frac{1}{3} \int_{-1}^2 f(x) dx = -4 \Rightarrow \int_{-1}^2 f(x) dx = -12$$

$$\frac{1}{5} \int_2^7 f(x) dx = 8 \Rightarrow \int_2^7 f(x) dx = 40$$

Therefore the average value of f over the interval $[-1, 7]$ is

$$\frac{1}{7-(-1)} \int_{-1}^7 f(x) dx = \frac{1}{8} \left(\int_{-1}^2 f(x) dx + \int_2^7 f(x) dx \right) = \frac{1}{8} (-12 + 40) = \frac{28}{8} = \frac{7}{2}.$$

8-1

48.  Let f be the function given by $f(x) = (x^2 + x) \cos(5x)$. What is the average value of f on the closed interval $2 \leq x \leq 6$?

(A) -7.392

(B) -1.848 ✓

(C) 0.722

(D) 2.878

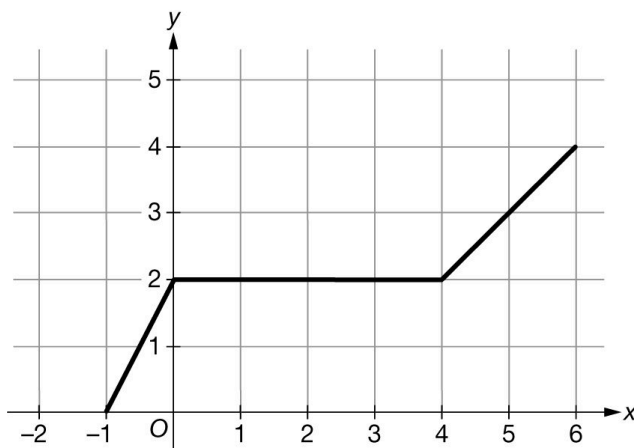
Answer B

Correct. The average value of a function f over the interval $a \leq x \leq b$ is $\frac{1}{b-a} \int_a^b f(x) dx$. Applying that to this problem gives the following calculation that can be done with the calculator.

$$\frac{1}{6-2} \int_2^6 (x^2 + x) \cos(5x) dx = -1.848$$

8-1

49.



The graph of the continuous function f consists of three line segments, as shown in the figure above. What is the average value of f on the interval $[-1, 6]$?

(A) $\frac{4}{7}$

(B) 2

(C) $\frac{15}{7}$

(D) 15

**Answer C**

Correct. The average value of a continuous function f over the interval $a \leq x \leq b$ is $\frac{1}{b-a} \int_a^b f(x) dx$. Applying that to this problem gives the following calculation.

$$\frac{1}{6-(-1)} \int_{-1}^6 f(x) dx = \frac{1}{7} \left[\frac{1}{2} (1)(2) + (4)(2) + \frac{1}{2} (2+4)(2) \right] = \frac{1}{7} [1 + 8 + 6] = \frac{15}{7}$$

8-1

50. Let f be a continuous function defined on the closed interval $2 \leq x \leq k$, where $k > 2$. If the average value of f on this interval is 6 and $\int_2^k f(x) \, dx = 42$, then $k =$

- (A) 9
(B) 8
(C) 7
(D) 5

**Answer A**

Correct. The average value of a continuous function f on the interval $[a, b]$ is $\frac{1}{b-a} \int_a^b f(x) \, dx$. On the interval $[2, k]$, the average value is 6. Therefore $\frac{1}{k-2} \int_2^k f(x) \, dx = 6 \Rightarrow \frac{1}{k-2} \cdot 42 = 6 \Rightarrow k = 9$.

51.  What is the average value of $y = e^{\sqrt{x}}$ on the interval $[1, \pi]$?

- (A) 1.479
(B) 4.246
(C) 4.302
(D) 9.092

**Answer B**

Correct. The average value of a function f over an interval $[a, b]$ is $\frac{1}{b-a} \int_a^b f(x) \, dx$.

$$\frac{1}{\pi-1} \int_1^{\pi} e^{\sqrt{x}} \, dx = 4.246$$

8-1

52. If $f(x) = \sqrt{x}$, what is the average value of f on the interval $[0, t]$ in terms of t ?

(A) $\frac{\sqrt{t}}{t}$

(B) $\frac{\sqrt{t}}{2}$

(C) $\frac{2\sqrt{t}}{3}$ ✓

(D) $\frac{2t\sqrt{t}}{3}$

Answer C

Correct. The average value of a function on an interval $[a, b]$ is $\frac{1}{b-a} \int_a^b f(x) dx$.

$$\frac{1}{t-0} \int_0^t \sqrt{x} dx = \frac{1}{t} \cdot \frac{2}{3} x^{\frac{3}{2}} \Big|_{x=0}^{x=t} = \frac{1}{t} \cdot \left(\frac{2}{3} t^{\frac{3}{2}} - 0 \right) = \frac{2}{3} \frac{t^{\frac{3}{2}}}{t} = \frac{2}{3} t^{\frac{1}{2}} = \frac{2}{3} \sqrt{t}$$

53.  For $t \geq 0$ hours, the velocity of a ship is modeled by $v(t) = \frac{1530}{t^2 - 24t + 172}$ miles per hour. What is the average velocity of the ship, in miles per hour, from $t = 1$ to $t = 15$ hours?

(A) 2.220

(B) 25.810

(C) 31.578

(D) 33.834 ✓

Answer D

Correct. The average value of a function f over an interval $[a, b]$ is $\frac{1}{b-a} \int_a^b f(x) dx$. Therefore, the average

velocity of the ship from $t = 1$ to $t = 15$ is $\frac{1}{15-1} \int_1^{15} v(t) dt = \frac{1}{14} \int_1^{15} \frac{1530}{t^2 - 24t + 172} dt = 33.834$.

8-1

54. The rate at which people register for a new social networking site is given by $g(t)$, where t is measured in minutes and $g(t)$ is measured in people per minute. The average rate of registration during the first 60 minutes is equal to the rate of registration at time $t = 30$ minutes.

Which of the following equations describes the situation?

(A) $\int_0^{60} g(t) dt = g'(30)$

(B) $\int_0^{60} g'(t) dt = g'(30)$

(C) $\frac{1}{60} \int_0^{60} g(t) dt = g(30)$



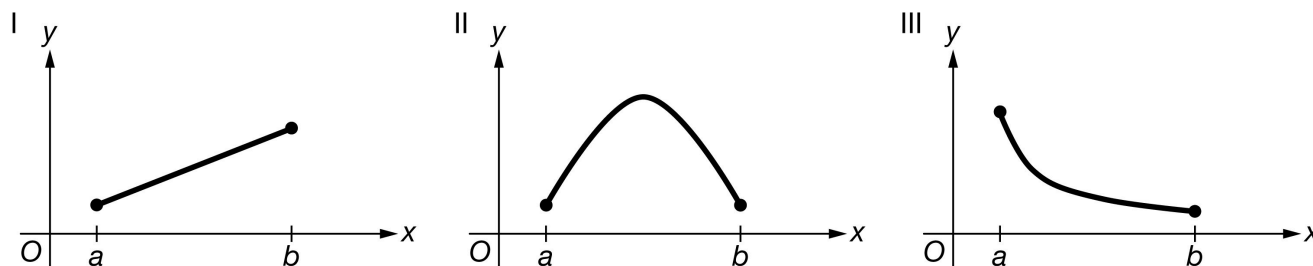
(D) $\frac{1}{60} \int_0^{60} g'(t) dt = g(30)$

Answer C

Correct. The average value of a function g over the interval $[a, b]$ is $\frac{1}{b-a} \int_a^b g(t) dt$. The average rate of registration during the first 60 minutes is thus $\frac{1}{60} \int_0^{60} g(t) dt$. The rate of registration at time $t = 30$ is $g(30)$. Therefore the equation $\frac{1}{60} \int_0^{60} g(t) dt = g(30)$ describes this situation.

8-1

55.



Let f be a continuous function that satisfies the condition $\frac{1}{b-a} \int_a^b f(x) \, dx = f(c)$ for exactly one value of c , where $a < c < b$. Which of the graphs above could be the graph of $y = f(x)$?

(A) I only

(B) II only

(C) I and III only

(D) II and III only



Answer C

Correct. The expression $\frac{1}{b-a} \int_a^b f(x) \, dx$ is the average value of the function f on the interval $[a, b]$. The average value will be between the absolute minimum and the absolute maximum of f on the interval $[a, b]$. The condition that $\frac{1}{b-a} \int_a^b f(x) \, dx = f(c)$ for exactly one value of c in the interval (a, b) means that the horizontal line $y = \frac{1}{b-a} \int_a^b f(x) \, dx$ will intersect the graph of f exactly once on the interval (a, b) . Only graphs I and III have that property and therefore could be the graph of $y = f(x)$. There would be two intersection points for graph II.